THISUL HERATH

ARTIFICIAL INTELLIGENCE UNDERGRADUATE

CONTACT

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PROFESSIONAL SUMMARY

Information Technology undergraduate specializing in Artificial Intelligence at SLIIT, with an overall GPA of 3.84 and hands-on project experience in machine learning, MLOps, and full-stack development. Led academic teams delivering retail forecasting, solar planning, and mobile collaboration platforms. Skilled in Python, scikit-learn, FastAPI, MLflow, React, and Node.js, with experience building reproducible ML pipelines, containerized services, secure REST APIs, and CI/CD workflows. Seeking an internship in AI, machine learning, or software engineering to apply these skills to practical problems and contribute to reliable software solutions.

EDUCATION

2024 - Present
BSc (Hons) in Information Technology Specializing in Artificial Intelligence (Current GPA - 3.84)
SLIIT | 2024 – 2028 | Expected graduation: 2028

2023/24
GCE Advanced Level (A/L) - Physical Science Stream
Royal College, Colombo 07.

2020/21
GCE Ordinary Level (O/L)
Royal College ,Colombo 07.

SKILLS

- Languages: Python, JavaScript, TypeScript, Java, SQL, C#, Dart
- AI / Machine Learning: Scikit-Learn, MLflow, NLTK, Imbalanced-Learn, Time-Series Forecasting, NLP Pipelines, LangGraph, rule-based agent workflows
- Frameworks & Libraries: React Native, React.js (Vite), Next.js, Node.js, Express.js, FastAPI, Spring Boot (MVC, REST), ASP.NET Core 8, Entity Framework Core, Flutter
- Databases & Cloud: MongoDB (Mongoose), Supabase (PostgreSQL), Firebase (Firestore), MySQL, Hibernate/JPA
- Tools & DevOps: Docker, GitHub Actions (CI/CD), Git, Expo / Expo Router, Jest, Pytest, Postman, Maven, Tailwind CSS, Figma
- Backend & Security: REST APIs, JWT Authentication, Role-Based Access Control (RBAC), Email OTP Verification

PROJECTS

2026
C# | ASP.NET CORE | REACT | FLUTTER | PYTHON | LANGGRAPH | POSTGRESQL
Link

Smart Solar Installation & Grid Compliance Platform (Group Project - Team Lead)

- Led a four-member team developing a rooftop solar planning platform covering customer surveys, field inspections, engineering approval, and equipment procurement.
- Combined React and Flutter applications with an ASP.NET Core API and PostgreSQL database, supporting JWT authentication and role-based access control.
- Used FastAPI and rule-based LangGraph workflows for preliminary solar sizing, grid-compliance screening, safety checks, and equipment pricing through a live exchange-rate API.
- Supported approval audit trails and transactional inventory reservations.

2026
PYTHON | FASTAPI | SCIKIT-LEARN | MLFLOW | DOCKER | GITHUB ACTIONS
Link

Churn Prediction Service (Individual Project)

- Built a customer churn prediction pipeline using the Kaggle Telco dataset, covering data preparation, feature engineering, and model training.
- Used MLflow to track experiments, model artifacts, and metadata for reproducibility.
- Developed a FastAPI inference service with input preprocessing and feature alignment to match the trained model's expected schema.
- Containerized the service with Docker and automated image building and publishing to Docker Hub through GitHub Actions.

2026
REACT | TAILWIND CSS| NODE.JS | SUPABASE| PYTHON | SCIKIT-LEARN
Link

7 Super City: Intelligent Retail & Forecasting Platform (Group Project - Team Lead)

- Led development of a retail management platform integrating POS, inventory tracking, customer loyalty analytics, and sales forecasting.
- Combined a React frontend with an Express.js API and a separate Python forecasting service to separate transactional operations from machine learning workloads.
- Integrated scikit-learn models for sales forecasting and Supabase PostgreSQL for managing products, inventory, customers, and sales records.
- Implemented JWT authentication, bcrypt password hashing, role-based access control, and database row-level security.

2026
REACT NATIVE | EXPO | TYPESCRIPT | NODE.JS
Link

UniVault: Academic Collaborative Resource Platform (Group Project - Team Lead)

- Led development of a cross-platform mobile application for sharing study materials, managing public and private collections, and organizing collaborative study groups.
- Built the mobile frontend using React Native, TypeScript, and Expo Router with file-based navigation.
- Developed a Node.js and Express.js backend with Multer for document and image uploads, using seven MongoDB models with compound text indexing and schema hooks.
- Integrated JWT authentication through React Context and centralized backend error handling.

2025
PYTHON | SCIKIT-LEARN NLTK | JUPYTER
Link

Automated NLP Text Preprocessing & Classification Pipeline (Group Project - Team Lead)

- Led development of a reproducible machine learning pipeline for preprocessing and classifying unstructured text.
- Applied text preprocessing, TF-IDF vectorization, and PCA dimensionality reduction to prepare features for model training.
- Addressed class imbalance using RandomOverSampler.
- Trained and tuned SVM and LinearSVC classifiers using GridSearchCV for hyperparameter optimization.

CERTIFICATES

Certificate in AI/ML Engineer - SLIIT

- Stage 1 - https://code.sliit.org/certificates/nddgq8tgtd
- Stage 2 - https://code.sliit.org/certificates/cbxmlxyplk

LEADERSHIP AND ACHIEVEMENTS

Athletic Achievements & Colors

- Western Province Colors: Awarded for Badminton (2022, 2023) in recognition of provincial-level excellence
- School Colors (Royal College): Triple recipient of School Colors for Badminton (2019, 2022, 2023).
- House Captain (Badminton): Appointed as House Captain in 2023, overseeing team strategy and player development
- Vice-Captain : Served as Vice Captain of the Royal College Under-15 and Under-17 Badminton team (2018, 2020).

Leadership & Service

- Junior Prefect: Appointed as a Junior Prefect at Royal College (2018), responsible for maintaining student discipline and assisting in school-wide event management.

REFERENCES

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